

SAATHI

Human-Centered Multimodal AI Response Agent

ONE CONVERSATION. ANY SITUATION. SAFE, AUTHORIZED, REAL-WORLD ACTION.

1. The Problem

When people face a real-world problem, the hardest part is often not getting an answer—it is knowing what to do next. A person may not know which hospital, government department, emergency service, route, form, or support channel is relevant. Existing chatbots mainly provide information; they do not reliably convert human intent into a verified action.

SAATHI addresses this gap by creating a single human interface to complex societal infrastructure.

2. Our Solution

SAATHI is not another AI chatbot. It is a multimodal, agentic response platform through which users can speak naturally, show an image, or provide text. SAATHI understands the situation, assesses risk, selects the right specialist capability, asks for authorization when needed, coordinates the appropriate service, executes the next action, and follows up until the case is resolved.

Human	Understand	Assess	Authorize	Act	Verify	Follow up	Resolve
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3. Why SAATHI Is Different

Human-first	Natural conversation; no need to understand departments, apps or procedures.
Action-oriented	Moves from intent to a concrete, trackable next step instead of stopping at an answer.
Safety-first	Deterministic risk policies, approval gates and a dedicated safety layer protect consequential actions.
Multimodal	Voice, text and camera/vision support English, Hindi and Telugu initially.
Follow-through	Cases continue after the first action through status tracking, notifications and follow-up.

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core workflows	normal agent target	target unsafe action rate	unified interface

Target users: citizens, elderly people, people with accessibility needs, students/young adults, and authorized family caregivers.

THE PROPOSED SYSTEM

Four high-impact workflows powered by one agentic platform

01 MEDICAL COORDINATION

Voice → risk assessment → nearby hospital discovery → user authorization → simulated contact → case tracking → follow-up.

Boundary: SAATHI supports triage and coordination; it does not diagnose disease.

02 DISASTER RESPONSE

Voice/location → local risk estimation → affected-zone assessment → safe route/shelter discovery → personalized response.

Prototype risk model can use rainfall, temperature, elevation, drainage, historical incidents, roads and population.

03 CIVIC ASSISTANCE

Voice → transcription/language detection → intent & entity extraction → department classification → complaint draft → structured case → tracking.

Turns an everyday complaint into an actionable, traceable case.

04 ACCESSIBILITY & VISION

Camera → OCR/vision → scene or document understanding → simplified explanation → spoken response.

Designed to reduce information barriers through multimodal and multilingual interaction.

5. Technology & Architecture

Experience: React Native + Expo mobile app, Next.js + TypeScript web dashboard. Backend: NestJS API with Python/FastAPI AI/ML services. AI: reasoning model + structured tool workflows, specialist agents and deterministic safety policies. Voice: real-time speech-to-speech with WebRTC. Data: PostgreSQL + pgvector, Redis; event-driven workflows. ML: PyTorch, scikit-learn/XGBoost, evaluation and safety benchmarks.

Voice / Text / Camera	Multimodal Gateway	Intent + Risk	Agent Orchestrator	Services / Tools	Approval Gate	Action + Case	Follow-up
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6. Trust, Safety & Human Control

- Sensitive actions require explicit authorization; every tool declares risk, permissions, data shared and audit requirements.
- Emergency handling uses a governed safety policy rather than allowing an LLM to make unrestricted critical decisions.
- Structured AI outputs are validated server-side; critical actions are never triggered by free-form model text.
- Privacy follows data minimization, purpose limitation, consent, retention controls and auditability.
- The system never claims an action succeeded unless a verified tool result confirms it; simulated integrations are clearly labeled.

JURY PROPOSAL

What we will demonstrate, why it matters, and how it scales

7. The Jury Demo

Scene 1	A user describes an urgent health concern. SAATHI assesses risk, identifies a suitable hospital, requests authorization, creates the case and shows follow-up.
Scene 2	A user reports flood concern. SAATHI uses location and the disaster-risk workflow to show risk and a safer route/shelter option.
Scene 3	A civic complaint is spoken naturally. SAATHI classifies it, structures the case and demonstrates tracking.
Scene 4	A user points the camera at a document. SAATHI understands it and explains the content through voice.

8. Innovation & Impact

- One interface, many domains: medical, disaster, civic and accessibility use cases are unified by the same response loop.
- From conversation to outcome: SAATHI treats every real-world issue as a Case with status, actions, authorization and resolution state.
- Trust as infrastructure: consent, safety policies, audit trails and verifiable tool outcomes are part of the core product—not add-ons.
- Designed for inclusion: voice-first interaction and multilingual multimodal support reduce dependence on technical literacy.
- Scalable foundation: future agents can extend the platform to education, employment, public services, financial safety and family assistance.

9. MVP & Roadmap

Phase 1 — TechFusion 2026	Four polished workflows; voice-first experience; agent routing; risk engine; tool calls; authorization; case management; follow-up; audit trail. Simulated external integrations are explicitly labeled.
Phase 2 — Pilot	Pilot with one hospital, one NGO, one municipal body and one emergency workflow; validate safety, usability and outcome metrics.
Phase 3 — Regional	Expand service integrations, multilingual capability, operational dashboards and verified provider networks.
Phase 4-5 — Scale	Government/health integrations followed by broader regional and nation-scale deployment, subject to legal, privacy, safety and provider-readiness gates.

10. Success Measures

User Task completion • time-to-action • satisfaction	AI Intent/risk accuracy • hallucination • tool selection	Safety Unsafe action • unauthorized sharing • escalation errors	Societal Resolved cases • coordination time • people assisted
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“We are not building another AI chatbot. We are building an AI layer between humans and the services they need.”

Team Meta Minds	Ownership
Mohd Junaid Pasha — Team Leader	Product direction, architecture, orchestration, demo, integration

Muhammad Saif Patel — AI/ML	ML models, evaluation, agent intelligence, risk engine
Mustyala Shiva Sai Tej — Full-Stack / Systems	Backend, database, workflows, APIs, deployment